

Non-monic quadratic trinomials



- 1
- a $8x^2 + 11x + 3 = (8x + 3)(x + 1)$
- b $2x^2 - 19x + 45 = (2x - 9)(x - 5)$
- c $2x^2 + 3x - 20 = (2x - 5)(x + 4)$
- d $10x^2 + 29x + 21 = (2x + 3)(5x + 7)$

- 2
- a $(2x + 5)(x - 8)$
- b $(3y + 1)(y + 9)$
- c $(3t - 4)(4t + 1)$
- d $(9x + 4)^2$
- e $(8x - 3)^2$
- f $(3x - 4)(x - 7)$
- g $(8x - 3)(x - 2)$
- h $(8x + 3)(x - 3)$
- i $(4x + 5)(3x - 2)$
- j $(-6x + 7)(x + 8)$
- k $(-2x + 3)(3x + 2)$
- l $-(2x - 7)(3x - 2)$

- 3
- a $3(x - 9)(x + 2)$
- b $4(x + 5)^2$
- c $4(x + 2)(x + 4)$
- d $5(2x - 3)(x + 2)$
- e $2(s + 3)(4s - 9)$
- f $4(x - 9)(x + 8)$
- g $2x(x + 5)(x + 3)$
- h $-4(x - 5)(x + 2)$
- i $3(x - 5)(x - 2)$
- j $-3(x - 2)^2$

4 $k = 9$

Applications

- 5 $(4 + x)(3 - x)$ units²
- 6 $2x + 5, 3x + 4$
- 7
- a $10x^2 + 105x + 245$
- b $5(2x + 7)(x + 7)$
- 8
- a $10y^2 - 112y + 22$
- b $2(5y - 1)(y - 11)$
- 9 $(3x + 1)(x + 6)$

10

a $6(x+3)^2$



b $(x+3)$ units

11 $m = -6, n = -8$

12

a $(4x^2 - \pi x^2)$ units²

b $x^2(4 - \pi)$

13 $-10(t+2)(t-7)$

14

a $-16(t+1)(t-2)$

b 0

c 0

d 2 seconds after jumping off the board, Tara hits the water.